

Day 23 – AWS Data Engineering with Glue, Athena & Redshift

Introduction

Data Engineering focuses on collecting, transforming, storing, and analyzing data.

AWS Glue

Serverless ETL service for data cleaning and transformation.

Amazon Athena

Serverless SQL query service for S3 datasets.

Amazon Redshift

Cloud data warehouse for analytics.

Amazon S3

Scalable data lake storage service.

Workflow

Raw data → ETL → Queries → Analytics dashboard.

Real-World Project

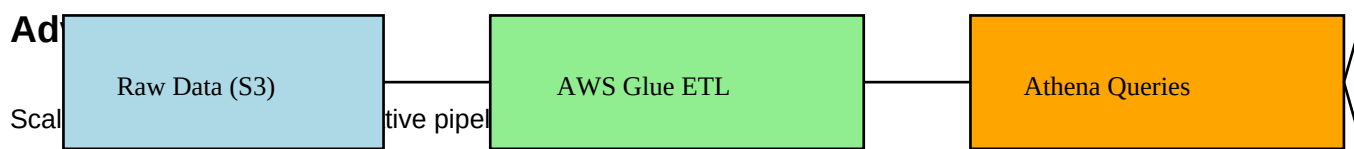
Student Analytics Platform using AWS services.

Security

IAM Roles, encryption, monitoring, secure access.

Monitoring

CloudWatch monitors ETL jobs and analytics performance.



Use Cases

BI dashboards, AI pipelines, enterprise analytics.

Data Engineering Architecture Diagram

AWS Data Engineering Services

AWS Service	Purpose
AWS Glue	ETL & transformation
Athena	Serverless SQL analytics
Redshift	Data warehouse
S3	Data lake storage
CloudWatch	Monitoring

Author: Sankar S – Cloud & AI Learning Journey